Practice workbook · Writing a sigma sum as a formula in n

Recognising that a sum like the sum of (ar + b) from r = 1 to n is arithmetic and writing it as an expression in n.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-04

1. Question 3 · part a · 3 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part a only.

3 (a) Show that $\sum_{r=1}^{n}(5r-3)=\frac{n}{2}(5n-1)$ (3)

(b) Hence, or otherwise, evaluate $\sum_{r=31}^{60}(5r-3)$ (2)

Given that $\sum_{r=1}^{n}(5r-3)=3783$

(c) find the value of n (3)

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1. Question 3 · part a · 3 marks · Working and answer

Working and answer

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2. Question 6 · part a · 3 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

6 (a) Show that $\sum_{r=1}^{n}(4r-3) = n(2n-1)$

(3)

(b) Hence, or otherwise, find the least value of n such that $\sum_{r=1}^{n}(4r-3) > 1000$

(3)

Given that $S_n = n(2n-1)$, $t_n = (4n-3)$ and that $18 + 3t_{n+7} = S_{n+4}$

(c) find the value of n.

(4)

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2. Question 6 · part a · 3 marks · Working and answer

Working and answer

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3. Question 5 · part a · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

5 (a) Show that $\sum_{r=1}^{n}(3r+5) = \frac{1}{2}n(3n+13)$ (3)

(b) Hence evaluate $\sum_{r=35}^{50}(3r+5)$ (2)

Given that $\sum_{r=1}^{n}(3r+5) = 385$

(c) find the value of n. (3)

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P 6 2 2 8 5 A 0 1 0 3 2

3. Question 5 · part a · 3 marks · Working and answer

Working and answer

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4. Question 1 · part a · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part a only.

1 (a) Show that $\sum_{r=1}^{n}(3r+2) = \frac{n}{2}(3n+7)$

(3)

(b) Hence, or otherwise, evaluate $\sum_{r=10}^{40}(3r+2)$

(2)

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(Total for Question 1 is 5 marks)

4. Question 1 · part a · 3 marks · Working and answer

Working and answer

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5. Question 9 · part a · 3 marks

2021 NOV · Paper 01

Further practice · Attempt part a only.

9 (a) Show that $\sum_{r=1}^{n}(5r-1)=\frac{n}{2}(3+5n)$ (3)

(b) Hence, or otherwise, evaluate $\sum_{r=10}^{20}(5r-1)$ (3)

The sum of the first n terms of an arithmetic series is S_n where $S_n=\sum_{r=1}^{n}(5r-1)$

The rth term of this series is u_r

Given that $S_n=12u_{n+1}+52$

(c) find the value of n (5)

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5. Question 9 · part a · 3 marks · Working and answer

Working and answer

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6. Question 6 · part b · 6 marks

2022 JAN · Paper 02

Further practice · Attempt part b only.

6 The sum of the first n terms of an arithmetic series is S_n where $S_n = n(3 + 2n)$

(a) Find the value of S_{20}

(2)

Given that $S_n = \sum_{r=1}^{n} (Ar + B)$

(b) find the value of A and the value of B

(6)

A different arithmetic series has first term 7 and common difference 4

The sum of the first n terms of this series is T_n

(c) Use algebra to find the value of n for which $T_n = S_n + 252$

(5)

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6. Question 6 · part b · 6 marks · Working and answer

Working and answer

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7. Question 4 · part a · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part a only.

4 Given that $\sum_{r=1}^{n}(4r+A)=\frac{n}{2}(4n-6)$ where A is a constant

(a) show that $A=-5$

(3)

Hence, or otherwise,

(b) (i) evaluate $\sum_{21}^{60}(4r-5)$

(2)

(ii) find the greatest value of n such that

$$\sum_{r=1}^{n}(4r-5)<3418$$

(3)

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P 7 8 9 6 4 A 0 8 3 6

7. Question 4 · part a · 3 marks · Working and answer

Working and answer

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